A COMPLETELY UNIFORM NESTED STEINER QUADRUPLE SYSTEM OF ORDER 56

Abstract. A *nested Steiner quadruple system* of order v is a set of blocks $\{x, y \mid z, w\}$, each an unordered pair of disjoint pairs, whose underlying 4-sets form an SQS(v); it is *completely uniform* if every one of the $\binom{v}{2}$ pairs occurs as a nested pair and all of them occur equally often. In the e-print of Lu's census — the only text of it we could read — every admissible order v with $8 \le v \le 50$ is settled, and the census closes with the remark that "the next unresolved case is $v = 56$", expected there to be hard because the existence of a rotational SQS(56) is undetermined. We exhibit a completely uniform nested SQS(56). Its underlying design is obtained from Lu's SQS(14) by two Hanani doublings, so no rotational system is used, and the nesting is written out in full as one ternary digit per block: all 1540 pairs occur, each with multiplicity $9 = (56-2)/6$.

1. The source statement

Let V be a v-set. A *Steiner quadruple system* SQS(v) is a collection $\mathcal{B}$ of 4-subsets of V covering every 3-subset of V exactly once; such a system exists iff $v \equiv 2$ or $4 \pmod 6$, or $v \in \{0, 1\}$, and then

$$|\mathcal{B}| = b(v) = \frac{v(v-1)(v-2)}{24}. \tag{1}$$

Following Lu [3, §1], a *nested* SQS(v), written NSQS(v), is a pair $(V, \mathcal{N})$ in which each element of $\mathcal{N}$ is an unordered pair of disjoint pairs of V, written $\{x, y \mid z, w\}$ and called a *nested block*, such that the 4-sets $\{x, y, z, w\}$ obtained by forgetting the partition form an SQS(v). The two pairs $\{x, y\}$ and $\{z, w\}$ of a nested block are its *nested pairs*. Lu calls a nested SQS *uniform* if every pair that occurs as a nested pair occurs the same number of times, and records the following definition verbatim [3, §1]:

> "A uniform nested SQS is called *completely uniform* if, in addition, all pairs in $\binom{V}{2}$ appear as nested pairs."

Lu's general existence theorem for complete uniformity is confined to orders 2^m; §5 of [3] is a list of individual examples, closing with a table captioned *"Existence of completely uniform nested* SQS(v) *for* $8 \le v \le 50$*"*, listing all eight admissible orders $8, 14, 20, 26, 32, 38, 44, 50$ as *Exists*, followed immediately by this sentence:

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> "The next unresolved case is $v = 56$. Because the existence of a rotational SQS(56) is undetermined, this case is especially challenging and may be difficult to approach even with computer search."

Both quotations are taken from the e-print version v3 of [3] (9 November 2025), the only full text we could reach; the published journal text was not available to us, so a change made at proof stage — including a rewording or a closing of the $v = 56$ case — would be invisible here.

2. The arithmetic of order 56

Complete uniformity forces the multiplicity. Each nested block contributes exactly two nested pairs, so a completely uniform NSQS(v) of multiplicity μ satisfies $2b(v) = \mu\binom{v}{2}$, i.e.

$$\mu = \frac{2b(v)}{\binom{v}{2}} = \frac{v-2}{6}, \tag{2}$$

which is an integer exactly when $v \equiv 2 \pmod 6$ [3, §2], [1, §4]. At $v = 56$ everything is pinned:

$$\begin{gathered} b(56) = \frac{56 \cdot 55 \cdot 54}{24} = 6930, \qquad \binom{56}{2} = 1540, \\ \mu = \frac{2 \cdot 6930}{1540} = 9 = \frac{56-2}{6}. \end{gathered} \tag{3}$$

The orders $51, \ldots, 55$ are $3, 4, 5, 0, 1 \pmod 6$, so none of them is admissible for complete uniformity: 56 is the next admissible order after 50.

Theorem 1. *A completely uniform nested Steiner quadruple system of order* 56 *exists. Explicitly, the nested system $\mathcal{N}$ of §4 has* 6930 *nested blocks, its underlying* 4*-sets form an* SQS(56)*, all* 1540 *pairs of the point set occur as nested pairs, and every one of them occurs exactly* 9 *times.*

The proof occupies §3 and §4: §3 builds the underlying SQS(56) from data printed here, and §4 prints the nesting.

3. The underlying SQS(56)

Step 1: an SQS(14)**.** Identify $\mathbb{Z}_7 \times \{0, 1\}$ with $\{0, 1, \ldots, 13\}$ by $(x, y) \mapsto x + 7y$, and develop each of the thirteen 4-sets

$$\begin{gathered} \{0,1,10,11\},\ \{0,1,4,2\},\ \{0,2,9,7\},\ \{0,4,13,10\},\ \{0,7,4,11\}, \\ \{0,8,7,1\},\ \{0,8,13,2\},\ \{0,9,11,8\},\ \{0,9,1,5\},\ \{0,10,12,2\}, \\ \{0,12,9,4\},\ \{0,13,12,1\},\ \{7,13,12,10\} \end{gathered} \tag{4}$$

under $x \mapsto x + 1 \pmod 7$ acting on the first coordinate. The $13 \cdot 7 = 91$ translates are distinct and equal $b(14)$, and they cover each of the 364 triples of a 14-set exactly once, so they form an SQS(14). These are the underlying

4-sets of the thirteen base blocks tabulated in [3], with Lu's nested pairings discarded.

Step 2: a 1-factorization. For even n, let the round-robin 1-factorization of K_n on $\{0, \dots, n-1\}$ be

(5)
$$F_i = \{\{n-1, i\}\} \cup \Big\{ \{(i+j) \bmod (n-1),\ (i-j) \bmod (n-1)\} : 1 \le j \le \tfrac{n-2}{2} \Big\},$$

for $i = 0, \dots, n-2$; the vertex $n-1$ plays the role of infinity. Each F_i is a perfect matching and the $n-1$ of them partition $E(K_n)$.

Step 3: Hanani doubling. The following is Hanani's $v \to 2v$ construction [2], stated and proved in the normalisation used here.

Lemma 2. *Let $(X, \mathcal{B})$ be an* SQS(n) *with $X = \{0, \dots, n-1\}$, and let $F_0, \dots, F_{n-2}$ be a 1-factorization of K_n on X. Write $x_\ell := x + n\ell$ for $x \in X$ and $\ell \in \{0, 1\}$, so that $\{x_\ell\}$ is a labelling of $\{0, \dots, 2n-1\}$. Then the following 4-sets form an* SQS$(2n)$*:*

(a) *$\{x_\ell, y_\ell, z_\ell, w_\ell\}$ for every $\{x, y, z, w\} \in \mathcal{B}$ and every $\ell \in \{0, 1\}$;*
(b) *$\{x_0, y_0, x_1, y_1\}$ for every pair $\{x, y\} \subseteq X$;*
(c) *$\{x_0, y_0, z_1, w_1\}$ for every* ordered *pair $(\{x, y\}, \{z, w\})$ of distinct edges lying in a common 1-factor F_i.*

Their number is $2b(n) + \binom{n}{2} + (n-1)\frac{n}{2}\big(\frac{n}{2} - 1\big) = b(2n)$.

Proof. Split a triple of $\{0, \dots, 2n-1\}$ by how it meets the two levels.

A triple $\{x_0, y_0, z_0\}$ inside level 0 lies in no block of type (b) or (c), since those meet each level in exactly two points; and it lies in exactly one block of type (a), namely the level-0 copy of the unique block of $\mathcal{B}$ containing $\{x, y, z\}$. Symmetrically for level 1.

A triple $\{x_0, y_0, z_1\}$ with two points at level 0 lies in no block of type (a). Suppose first $z \in \{x, y\}$, say $z = x$. The type-(b) block $\{x_0, y_0, x_1, y_1\}$ contains it, and no other type-(b) block does, because a type-(b) block is determined by its level-0 pair. A type-(c) block containing it would need the two distinct edges $\{x, y\}$ and $\{x, w\}$ in a common 1-factor, which is impossible because distinct edges of a 1-factor are disjoint; so no type-(c) block contains it, and the triple lies in exactly one block. Now suppose $z \notin \{x, y\}$. No type-(b) block contains $\{x_0, y_0, z_1\}$, since the level-1 pair of a type-(b) block is the level-1 copy of its level-0 pair. The pair $\{x, y\}$ lies in exactly one 1-factor F_i; in F_i the point z is matched to a unique w, and $w \notin \{x, y\}$ because $\{x, y\} \in F_i$ is disjoint from the other edges of F_i. Hence $(\{x, y\}, \{z, w\})$ is the unique ordered pair of distinct edges of a common 1-factor giving a type-(c) block through $\{x_0, y_0, z_1\}$, and there is exactly one such block. The mirror case $\{x_1, y_1, z_0\}$ is covered by the reversed ordered pair $(\{z, w\}, \{x, y\})$; this is where part (c) requires *ordered* pairs. Every triple is therefore covered exactly once, and the count follows by (1). ☐

Applying Lemma 2 to the SQS(14) of Step 1 with the 1-factorization (5) of K_{14} gives an SQS(28) with

$$2 \cdot 91 + \binom{14}{2} + 13 \cdot 7 \cdot 6 = 182 + 91 + 546 = 819 = b(28)$$

blocks; applying it again, with the 1-factorization (5) of K_{28}, gives an SQS(56) with

$$2 \cdot 819 + \binom{28}{2} + 27 \cdot 14 \cdot 13 = 1638 + 378 + 4914 = 6930 = b(56) \tag{6}$$

blocks. Neither doubling uses any group action on the doubled point set, and in particular no rotational SQS(56) is used anywhere.

Step 4: the canonical order. Write each of these 6930 blocks as an increasing 4-tuple ($a < b < c < d$) and sort the resulting list lexicographically. Call the result $B_0, \dots, B_{6929}$; thus $B_0 = (0, 1, 2, 4)$ and $B_{6929} = (51, 53, 54, 55)$. Everything in §4 refers to this order, which is determined by (4), (5) and Lemma 2 alone.

4. The nesting

Given a block $B_i = (a, b, c, d)$, its three perfect matchings are indexed by a ternary digit

$$t = 0 \mapsto \{a, b \mid c, d\}, \qquad t = 1 \mapsto \{a, c \mid b, d\}, \qquad t = 2 \mapsto \{a, d \mid b, c\}, \tag{7}$$

so a nesting of the SQS(56) of §3 is precisely a vector $(t_0, \dots, t_{6929}) \in \{0, 1, 2\}^{6930}$. The witness is the following vector.

Encoding. Read $t_0, \dots, t_{6929}$ as the base-3 digits of a non-negative integer N with t_0 least significant, write N in big-endian order into 1373 bytes, and base-64 encode those bytes. The result is the following 1832 ASCII characters, being the concatenation of the 29 lines below with no separator of any kind (the final line carries the single `=` pad):

hF9sYBi/phDk3uDEIIRpc3YPCtCl0msS3gE9cj9VqFOQsji+JZGju+ceeQRUUCm3
+P17kIPeTVPgT5tNtC4meAhXExMYoxSZ+6ec/b+EUYi/c6yDYNCKAqDnlyW9S7Cc
YRevrI4uIXCJ9D4vuBzV2OsF17fFbFjYWUUIa3YKwL3wfhrvr+a0wytpJaHxQErw
7Lzv4i3Jk2BZUmRarFVQ5DV+XzlGIAY6ZG4WNlVZ8lvc7SV0Al+7AuVdk7kI5LgZ
Tv/hb64QI1oOFavaN2IVPtHVDlpTeVcllnPDuW+mrJm2JE8RlERrmojt2KLbR/Dm
bG4N2Vx92/cpHws1sA9vBh7GmRrvUPfJjkXBD5Q/ca9XKJsugiyrkPR6+QLwnUi7
qVoMBOC6NshTzvPfKsDK+nVWKQ5AbWFDAkoUH58ne32tsfvDbMAyNj54DbtxASTx
7OME+8+uie09QSSFfJjuIzYqGZg8tKm8LelnhqXPx4NeQNWw/1SGPkJYyIGm8r97
+KwPXKNniuK8kxcQ3PVyO6tHm7Kx8i8SXv9xFmf3mSiaRhV/U8TOw4Qg5g/DTA+i
p2yR8Bt422FTQKHFRi9FgCo8xZCs+gzeiVVF4Mk3wu0ng5bWb0x4gEdyTcTOIKvf
uNlyVPmlPQWdH1bQRXGz9aSxRQRPJXDT5PngSXDPI+kpnPqQoCIkz4JXV9XzjjAy
QfzONSPEiiy0XWaApN1fz4Q2NwqTHSnP9GpvkSF3AtrTOQ5+5+J2Kp4lfc+OZH0q
SNOe8l64HBwBqhjQ9yrKhGDIKTpj7faNRAnqmpTa7B5PAqfwsrPiV2GDTx9s4dVv
F7laoL0A8MX+LAwJqnTBobeIgiLWc5osbg+zWuVFi6y+cxf8tv9b/oTvDHxQS50k
TcIo4jxx1SJuSgP5UwpdQAl+JvAlai8gpToM0DcJQmoOBZEM3n8LJu0s0rYklFLv
TVu8H8auEagap+Nx7SJehkjoFRfT+qMYGMGinyLTc+k2OL6Oy39mvUnnNH2BaZdj
ypVAEZgl2S2X0uzWWIWib3uP8RyjaiP9kwGxtCR5EqeVwnj0QAK/AS4Me7bkZqnf
X8vFNKUv9ZEOWjB3sKZGT38HY4wF8V6Z/0ri+3O9bM0vZFkPcu9tgAIrmbg7aGkP
3j9ENOq8pM0XKsJaNAruYQkyEz/2pNuXDPGs7SDzyEEYan5wRT9Er5w9cqVBnONd

/Z21das/eqFg3fqcW+h9iuBVLoU8s4uDokE4SjN9HMdu1IXKwoiyIgBvuu9GN115
16u59LJp2GrtHV5Io5+XtH24pUDUKOrP+TJPkHHX2woE9JwNV9BMzaeulbTEh//0
ObalAxRcF3qWwrzH25zhklBemnW+fNVCJnkP2s9Y76pOOLISgHBc4QqcTbDu1Kxl
cDKxlHulkULfVn7OjWTDFSquZ23/jcwyeSrLQG6sUkHrzQYn+XKzHin+Scj5CGLT
zzuZLe2Q+Wby8JoWZExo3YimMAvOU+RFRi0AFatuxhwjQJ7c2O3UCc6F5suSJ7Wo
GjiLEcs4J/14Oq+5APHxfvHs/wgTTPNXVY+zSBOBH6TZB82QxeO5ROccbd4AFfJi
OLpNkVNj3CvmuE1T/GiS3p9tqoG4yi5BTMmCzSAs7rSWiTXNE5UZlkOOMrNZrX7J
PAR5cdXpYW5u5HWCyiorA3m5o/J4iIgtIByMOOhN967/aTW4ti8paEn0lKNqFYr0
sTARWLsgX+gFYudKk77xIoa9rEzemRIVpY/xbdfNtGavUguzQEDdabvnXpCQjPGC
CqJTpqzgP4gW0/KuKrI5mO92ezE8LFaBl3bkY3c=

Its SHA-256, taken over those 1832 ASCII bytes, is

f802277ca20184e1b026bddc89ed1b30ed791885fa832ef17b657fcc5b2970b8,

and the SHA-256 of the 6930 decoded digits, taken as 6930 raw bytes with values in $\{0,1,2\}$, is

e5a8368b4a0b9f21d4e8e44be375a1cc9e00a364804e2b2366fa04e60aaf4016.

Decoding terminates exactly: after extracting 6930 digits the remaining quotient is 0, so the encoded integer carries no digit above t_{6929}.

Transcription check. As a check on the base-64 string above, the first 200 digits $t_0, \ldots, t_{199}$ — and only those — are also printed in plain decimal, read left to right:

20110111202222101002110202021011002010221201210012021202220021000
10201012221121002112120000211102112120102202210102001221122122121
2220212210100110101000112212100122112120002020112200211220001120110000

(the three lines concatenate, with no separator, to the 200 digits). Applying (7) to $B_0, \ldots, B_{19}$ with the digits $t_0, \ldots, t_{19}$ gives the first twenty nested blocks:

$$\begin{gathered}
\{0,4 \mid 1,2\} \quad \{0,1 \mid 3,6\} \quad \{0,5 \mid 1,9\} \quad \{0,7 \mid 1,8\} \quad \{0,1 \mid 10,11\} \\
\{0,12 \mid 1,13\} \quad \{0,14 \mid 1,15\} \quad \{0,16 \mid 1,26\} \quad \{0,25 \mid 1,17\} \quad \{0,1 \mid 18,24\} \\
\{0,23 \mid 1,19\} \quad \{0,22 \mid 1,20\} \quad \{0,27 \mid 1,21\} \quad \{0,29 \mid 1,28\} \quad \{0,30 \mid 1,54\} \\
\{0,1 \mid 31,53\} \quad \{0,32 \mid 1,52\} \quad \{0,1 \mid 33,51\} \quad \{0,1 \mid 34,50\} \quad \{0,49 \mid 1,35\}.
\end{gathered}$$

Proof of Theorem 1. Let $\mathcal{N} = \{\, (7) \text{ applied to } B_i \text{ with } t_i \;:\; 0 \le i \le 6929 \,\}$, where $B_0, \ldots, B_{6929}$ is the canonical order of §3 and (t_i) is the vector decoded above. By Lemma 2 and Step 4, the underlying 4-sets of $\mathcal{N}$ form an SQS(56), and by (7) each nested block is an unordered pair of disjoint pairs whose union is its own block, so $\mathcal{N}$ is a nested SQS(56) with 6930 nested blocks. It remains to tabulate, for each of the 1540 pairs of the point set, how many of the 6930 nested blocks have it as a nested pair. Doing so gives every pair the multiplicity 9: all 1540 pairs occur, none is missing, and the multiplicities sum to $1540 \cdot 9 = 13\,860 = 2 \cdot 6930$, as (2) and (3) require. Hence $\mathcal{N}$ is completely uniform. □

The tabulation in the last paragraph is a single exact integer pass over data printed in this paper: rebuild the 6930 blocks from (4), (5) and Lemma 2, sort them, decode the base-64 string, and increment a counter per nested pair. It involves no search and no solver, and it is what the program of §6 does.

5. Scope

One statement is settled: a completely uniform nested SQS(56) exists, the case named in the closing sentence of [3, §5]. The order 56 is a gap rather than the boundary of knowledge; among the examples of [1] is one headed "A complete uniform nested rotational SQS(62)". The $v = 56$ row of [1] also lists uniform but not completely uniform nestings, with 924 nested pairs at multiplicity 15 and with 1260 at multiplicity 11, both marked there as having no known design; nothing here bears on either. The reason [3] gives for expecting difficulty is that the existence of a rotational SQS(56) — one with an automorphism of order 55 fixing a point — is undetermined; our construction uses none, and since we have not computed the automorphism group of our SQS(56), whether it happens to be rotational is unknown to us. We exhibit one witness and make no enumeration, nonexistence, isomorphism or minimality statement; the general existence question for $v \equiv 2 \pmod 6$ remains open. Finally, as noted in §1, we could not read the published version of [3], nor reviewing databases behind paywalls, so a prior record of this case there would be invisible to us.

6. Verification

The program `verify.py` accompanying this note re-derives the computational assertions above from the data printed above and nothing else; it establishes nothing about the literature, so the statements of §5 about what is and is not in print are not among the things it checks. It is Python 3.9+, uses only the standard library, reads no external file, and all of its arithmetic is exact integer arithmetic. It rebuilds the SQS(14) from (4), checks the 1-factorizations (5) of K_{14} and K_{28}, performs both doublings and checks at each stage that the blocks are distinct and cover every triple exactly once (364, then 3276, then 27 720); it verifies the counts (6), the two SHA-256 digests of §4, the exact termination of the base-3 decoding, the agreement of the base-64 string with the 200-digit decimal prefix, and the twenty nested blocks printed above; and it then tabulates the 1540 nested-pair multiplicities and confirms that all are 9. Its recorded run, in `verify.output.txt`, reports

```
VERDICT: ALL 31 CHECKS PASS
```

and closes with its own list of what it does not cover. That count of 31 includes control checks on other orders and on deliberately corrupted input, which bear on the $v = 56$ witness only indirectly; the checks enumerated above are the ones Theorem 1 rests on.

References

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[3] X.-N. Lu, *Completely (quasi-)uniform nested Boolean Steiner quadruple systems*, J. Combin. Des. **34** (2026), no. 3, 139–156. doi:10.1002/jcd.22016; arXiv:2509.06663. All quotations refer to the e-print, version v3, the only full text we could reach.